

## TECHNICAL LEADERSHIP & WORK EXPERIENCE PRESENTATION

# Maheshchandra Hegde

**Founder & CTO | Building AI-Driven 3D Asset Platforms & Spatial Computing Solutions @srushtilabs.com**

16+ years delivering scalable enterprise frontend systems, high-reliability clinical platforms, offline-first architectures, and generative 3D spatial tools.  
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### Scalable Enterprise Systems

Philips Healthcare (Cyient), Cisco Systems,  
enterprise web apps, micro-frontends

### Startup Founder & Tech Director

Technoyana (technoyana.in), Twitan.com  
(high-reliability sports OS), InnoBrik

### 3D Spatial & Generative AI

Voxelforge AI (srushtilabs.com/voxelforge),  
ayam3d, CAE flight simulation

### Architectural Heritage

Rooted in CAD & 3D visualization assisting  
architect father; 50+ 3D projects delivered

# Professional Background, Architectural Heritage & Technical Philosophy

## ARCHITECTURAL LINEAGE & PASSION FOR 3D

### Natural Convergence of Architecture and Computing

- **Early Foundation with Architect Father:** Growing up assisting my architect father in his studio provided my earliest foundation in drafting, spatial coordination, blueprints, CAD, and architectural drawings. This practical exposure sparked an enduring passion for 3D modeling, spatial thinking, and visualization.
- **Blending Engineering with Human Interface:** Pursued formal education with a Bachelor of Engineering in Electronics & Communication and a Master of Science in Computing (Robert Gordon University, UK), specializing in Human Interface Design & Development.
- **16+ Years Hands-on Leadership:** Combined rigorous software architecture with creative spatial technologies—spanning flight simulation visual databases (CAE), enterprise dashboards (Cisco), healthcare platforms (Philips), startup products (Technoyana, Twitan), and generative 3D AI (Voxelforge AI, ayam3d).

## CAREER SNAPSHOT & CORE COMPETENCIES

### • **Frontend & Scalable Architecture:**

React, Angular, TypeScript, Node.js, Web Components, Micro-frontends, LitElement, Design Systems.

### • **Cloud, Edge & Reliability:**

GCP, Firebase, AWS, offline-first architectures, IndexedDB/Service Workers, WebSockets, real-time data sync.

### • **Spatial 3D & Generative AI:**

AI-assisted 3D generation (Voxelforge AI), automated mesh retopology (ayam3d), WebGL, Three.js, CAD drafting, 3D walkthroughs.

### • **Product & Startup Leadership:**

Founder/Co-Founder across Technoyana, Twitan.com, SrushtiLabs, InnoBrik; product strategy, sprint governance, mentoring.

### • **Key Industry Tenures:**

Cyient (Philips Healthcare), UST Global (Cisco Systems), Ness Technologies, Moonraft Innovation Labs, ThoughtFocus, CAE Simulation.

# Healthcare Mission-Critical Systems: Philips ICCA (Cyient Limited)

## Philips ICCA

### Intellispace Critical Care & Anesthesia

**Role:**

Senior Technical Lead (React / NodeJS)

**Tenure:**

Nov 2023 – Mar 2025

**Company:**

Cyient Limited for Philips Healthcare, Bangalore

**Operational Reality:**

24/7 continuous ICU hospital environments; zero tolerance for UI freeze or missing patient vitals

**Key Focus:**

Scalable frontend architecture, high performance, strict healthcare standards compliance

### 1. High-Performance Clinical Telemetry & Zero Latency

Architected React-based frontend modules responsible for rendering complex clinical time-series charts, vitals, and medication administration workflows. Enforced strict memory management and component memoization to prevent garbage collection pauses during long-running bedside sessions.

### 2. Decoupled Modular Architecture & Maintainability

Deconstructed complex clinical workflows into reusable, modular component libraries. Created robust contract boundaries with backend REST and streaming APIs, accelerating multi-developer collaboration while eliminating regression risks across unrelated clinical modules.

### 3. Healthcare Regulatory & Security Standards Compliance

Ensured all developed UI modules strictly complied with medical software safety benchmarks, HIPAA data protection guidelines, role-based clinician access controls, and robust audit trails for all patient data modifications.

# Product Engineering & Edge Resilience: Technoyana & Twitan.com

## Technoyana & Twitan

[technoyana.in](https://technoyana.in) | [twitan.com](https://twitan.com)

### Technoyana (2021 – 2023):

Co-Founder & Director; delivered fintech apps, product discovery engines, web & cross-platform mobile apps (React Native / Ionic).

### Twitan.com (Apr 2025 – Present):

Founder & Product Design Lead; AI-driven sports management, bracket engines & live scoring (Shutlify badminton OS).

### Operational Reality:

Tournament venues with dead-zones, unstable cellular Wi-Fi, and real-time live match arbitration.

### Tech Stack:

React, Angular, Node.js, Firebase/GCP, PWA, IndexedDB, WebSockets.

## 1. Offline-First Resilience for Unstable Physical Venues

Architected an offline-first state machine using IndexedDB local storage and Service Workers for courtside officials. When Wi-Fi or cellular connections fluctuate, the app operates without interruption; all scores and events queue locally and sync with conflict-resolution algorithms upon reconnection.

## 2. Cloud Architecture, Firebase & Rapid Delivery Workflows

Set up scalable backend cloud services using GCP, Firebase, and Node.js microservices. Evaluated and optimized tech stacks for clients, drastically reducing delivery timelines for complex fintech and consumer applications while keeping cloud operating expenses lean.

## 3. Modular Athlete & Tournament Dashboards

Designed modular dashboards integrating live statistics, tournament bracket generation (knockout, round-robin), court scheduling engines, and analytical insights, recognized for exceptional user experience and responsiveness.

# Spatial AI & Generative 3D Pipelines: SrushtiLabs (Voxelforge AI) & ayam3d

## SRUSHTILABS · VOXELFORGE AI ([srushtilabs.com/voxelforge/](https://srushtilabs.com/voxelforge/))

### Founder & CTO, SrushtiLabs ([srushtilabs.com](https://srushtilabs.com))

- Current Headline: Founder & CTO | Building AI-driven 3D asset platforms and spatial computing solutions @srushtilabs.com
- Live Application: [srushtilabs.com/voxelforge/](https://srushtilabs.com/voxelforge/) — Developing AI-based workflows for generative 3D asset creation and modular bundle assembly.
- Modular Game-Ready Bundles: Generates optimized, low-poly modular 3D assets ready for immediate assembly in Unreal Engine, Unity, and real-time WebGL engines.
- Rapid Prototyping Pipeline: Bridges text/image prompts into 3D voxel geometry with automated UV unwrapping and texture generation, cutting 3D asset prototyping time from days to minutes.
- Lightweight In-Browser Rendering: Specifically engineered to deliver compact asset payloads that stream effortlessly into web browsers without requiring heavy GPU rendering workstations.

### ayam3d — Advanced Mesh Synthesis & Retopology

- Exploratory R&D into automated 3D mesh synthesis, intelligent quad retopology, and PBR texture generation.
- Converts complex, unorganized high-poly meshes into clean, lightweight geometry with preserved edge flow.
- Solves real-time visualization challenges by creating Level of Detail (LOD) hierarchies for seamless 3D streaming.

### Applied Value to Physical & Digital Infrastructure

- Bridging Digital Twins with Web Applications: Enables lightweight 3D asset representations of physical spaces, facilities, and structures without multi-gigabyte download bottlenecks.
- Practical Spatial Computing: Demonstrates deep technical competence in polygon reduction, coordinate systems, and automated 3D pipelines that directly benefit modern spatial workflows and digital engineering.

## 3D Visualizer Expert Pedigree: 50+ Projects, CAE Simulation & Cisco Systems

### Architectural 3D & Walkthroughs

#### 35+ Commercial & Residential Projects

- Working with Architect Father: Built foundational expertise interpreting 2D plans, elevations, and structural schematics into accurate 3D models.
- Photorealistic Visualizations: Produced photorealistic exterior and interior architectural renderings, lighting studies, and texture mappings.
- 3D Walkthrough Animations: Produced animated camera walkthroughs for real estate developers and commercial builders to visualize spaces prior to construction.
- CAD & Drafting Precision: Hands-on mastery with CAD drafting tools, layer management, and dimensional accuracy.

### CAE Simulation Technologies

#### Visual Database Developer (2007 – 2008)

- Flight Simulator Visual Databases: Created high-fidelity 3D model libraries for military and civil aircraft full-flight simulators.
- Real-Time Terrain Modeling: Built elevation terrain databases, runway markings, airport ground equipment, and navigation landmarks.
- Strict Polygon & Memory Budgets: Developed real-time 3D models requiring deterministic frame rates (60 FPS locked) with zero visual pop-in or simulation stutter.
- Sensor & Environmental Simulation: Modeled day/night transitions, weather effects, and infrared/night-vision sensor textures.

### Cisco Systems Dashboard

#### Facility Dashboard & Smart Campus

- Cisco Facility Dashboard (2009 – 2010): Designed and developed an interactive facility dashboard during full-time on-campus consulting at Cisco Bangalore.
- Physical Infrastructure Telemetry: Visualized data center rack layouts, campus power consumption, cooling metrics, and environmental sensor telemetry.
- Awarded Cisco Appreciation: Received formal recognition from Cisco leadership in 2010 for delivering intuitive, high-impact data visualization.
- Continued Cisco Collaboration: Managed multiple Angular dashboards and UI applications for Cisco Systems via UST Global (2015 – 2019).

# Core Responsibilities & Technical Leadership Across 16+ Years

## 1. Scalable Architecture & System Design

- Architected enterprise frontend applications using React, Angular, and LitElement web components.
- Designed cross-platform UI Design Systems adopted across diverse development teams (Moonraft / TLC Hotels).
- Established robust contract boundaries using REST, WebSockets, and event-driven data streaming.
- Evaluated and selected modern cloud and web tech stacks for rapid, scalable client product delivery.

## 2. Operational Resilience & High Reliability

- Architected zero-loss offline-first caching mechanisms (IndexedDB/Service Workers) for field use (Twitan).
- Engineered life-critical clinical UI rendering with zero memory leaks for 24/7 hospital ICU units (Philips ICCA).
- Supervised cloud deployments on GCP, Firebase, and AWS with automated scaling and log monitoring.
- Designed real-time fallback pipelines ensuring continuous operational availability during network drops.

## 3. Spatial Computing & 3D Asset Pipelines

- Developed AI-assisted workflows (Voxelforge AI) for modular low-poly 3D asset generation (Unreal / Unity / WebGL).
- Directed R&D in automated 3D mesh retopology and PBR texture generation for lightweight web streaming (ayam3d).
- Created real-time 3D flight simulator databases with strict polygon and frame rate budgets (CAE).
- Built interactive 3D facility and infrastructure monitoring dashboards (Cisco Systems).

## 4. Agile Leadership, Mentorship & Delivery

- Led, mentored, and inspired cross-functional engineering squads of developers, UX designers, and QA analysts.
- Instituted design-to-code pipelines involving users early through rapid prototypes and usability feedback.
- Managed client deliverables, stakeholder communication, sprint planning, and engineering velocity.
- Recipient of multiple corporate excellence awards at UST Global and Cisco for leadership and client commitment.

## Major Achievements, Proven Track Record & Formal Acknowledgements

### 16+ Years

#### Professional Track Record

Extensive leadership across enterprise software, healthcare, startups & 3D tech

### 50+ Projects

#### 3D & Visualization Portfolio

Delivered across architectural visualization, CAD, flight simulation & spatial AI

### 4 Formal Awards

#### Industry Acknowledgements

Cisco Facility Appreciation + 3 UST Global Excellence & Client-First awards

### 100% Offline

#### Data Integrity at Edge

Zero data loss during field network drops with transactional replay queues

#### ★ Philips Intellispace Critical Care & Anesthesia (Cyient)

Led the frontend technical development of clinical modules for Philips Healthcare, delivering life-critical patient monitoring with rock-solid stability, zero memory leaks, and full healthcare regulatory compliance.

#### ★ Startup Co-Founder & Director (Technoyana & Twitan)

Successfully co-founded and steered Technoyana Digital Transformation Services and Twitan.com; built and shipped commercial fintech apps, product search engines, and the Shutlify badminton tournament operating system.

#### ★ Cross-Platform UI Design System (Moonraft Innovation Labs)

Architected a cross-platform UI design system using the LitElement web components library, creating a reusable component standard leveraged seamlessly across React, Angular, and Polymer teams for TLC Hotels.

#### ★ Formal Corporate Awards & Cisco Recognition

Received Cisco Systems Appreciation for developing the campus Facility Dashboard (2010); recognized with 3 Certificates of Appreciation at UST Global ('Inspiring People' 2018, 'Putting Client First' 2016, 'Living the Values' 2015).

## Real-World Technical Challenges Faced & Pragmatic Resolutions

### Challenge 1: Unstable Network Connectivity in Physical Venues (Twitan / Technoyana)

Operating live tournament arbitration in venues with crowded RF environments, intermittent Wi-Fi drops, and mobile dead-zones caused transaction failures and loss of live scores.

### Resolution Implemented:

Architected an 'Offline-First' state engine using IndexedDB client persistence and service workers. Implemented optimistic UI updates and a transactional event replay queue that automatically resynchronizes once the network reconnects.

**Result: Impact: 100% data preservation, zero operator workflow interruptions, seamless user experience.**

### Challenge 2: Clinical UI Performance & Memory Leaks in 24/7 ICU Monitoring (Philips ICCA)

Bedside patient monitors run uninterrupted for weeks. Complex real-time streaming vitals and continuous DOM updates risked browser memory accumulation and UI lag during critical clinical events.

### Resolution Implemented:

Enforced strict immutable state updates, decoupled rendering loops from data ingestion, implemented virtualized list rendering, and utilized Chrome DevTools heap profiling to systematically eliminate closure leaks.

**Result: Impact: Guaranteed stable 60 FPS UI rendering and flat memory profiles across multi-day continuous ICU sessions.**

### Challenge 3: High 3D Asset Payload & Browser Lag in Web-Based 3D (SrushtiLabs / Voxelforge AI)

Delivering detailed 3D models over the web created large download sizes, long initialization times, and frame drops on lower-spec client laptops and mobile devices.

### Resolution Implemented:

Developed automated retopology workflows (ayam3d) to decimate unnecessary polygons while preserving silhouette geometry; baked normal maps and bundled assets into lightweight modular formats for WebGL.

**Result: Impact: Reduced 3D model payload sizes by up to 75%, enabling instant in-browser loading and smooth real-time manipulation.**

# Lessons Learned & Core Engineering Best Practices

## 1. Involve Users Early Through Rapid Prototypes & Iterative Feedback

From my days establishing design-to-code pipelines at ThoughtFocus to architecting clinical modules at Cyient, building early interactive prototypes with real users uncovers fundamental workflow flaws before expensive engineering code is written.

## 2. Design for Real-World Edge Conditions, Not Ideal Lab Environments

Software rarely operates in perfect network and hardware conditions. Building applications with offline caching, graceful degradation, and asynchronous retry logic ensures systems remain reliable whether on a hospital floor, sports arena, or remote field office.

## 3. Standardize Design Systems & Component Libraries to Prevent Technical Debt

Re-inventing UI components across teams leads to inconsistent user experiences and bloated maintenance. Investing upfront in unified, framework-agnostic design systems (as demonstrated at Moonraft with LitElement) dramatically accelerates organizational velocity.

## 4. Ground Technology Decisions in Concrete Business & Human Impact

Technology is a vehicle for solving human problems. Whether building spatial AI tools (Voxelforge AI), sports arbitration engines (Twitan), or ICU monitoring (Philips), the ultimate metric of success is operational simplicity, reliability, and tangible value to the user.

## VISION FOR FUTURE PROCESS IMPROVEMENTS & SCALE

# Leveraging Proven Expertise in Scalable Systems, UX & Spatial Tech

- **Unified Spatial & Web Integration:**

Bridge complex 3D CAD, architectural models, and spatial data into lightweight, interactive web formats using progressive LOD and automated retopology (leveraging Voxelforge AI / ayam3d principles).

- **Resilient Edge-First Mobile Workflows:**

Implement offline-first client persistence and background transactional queuing across field-facing mobile applications, ensuring zero work loss regardless of remote connectivity challenges.

- **Standardized Enterprise Design Systems:**

Establish centralized, cross-platform component libraries and design tokens to streamline multi-team digital product delivery, enforce visual consistency, and eliminate duplicated engineering effort.

- **AI-Assisted Prototyping & Development:**

Incorporate generative AI and automation tools into early product discovery and asset creation pipelines, cutting design-to-code iteration cycles by 40%.

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